BEE 208 — Question 6: Transformer Ratio Calculator Menu

Control Statement: switch | Faculty of Engineering, ATU

1. ALGORITHM

Step 1: Start

Step 2: Declare variables choice (int), primaryTurns, secondaryTurns, primaryVoltage, secondaryVoltage, turnsRatio (double)

Step 3: Display transformer calculator menu (options 1 to 3)

Step 4: Read choice

Step 5: Use switch on choice

Step 6: Case 1 — read N1, N2; turnsRatio = N1 / N2; display turnsRatio

Step 7: Case 2 — read V1, N1, N2; V2 = V1 × N2 / N1; display V2

Step 8: Case 3 — read V2, N1, N2; V1 = V2 × N1 / N2; display V1

Step 9: Default — display 'Invalid choice'

Step 10: End

2. PSEUDOCODE

```
BEGIN
DECLARE choice AS INTEGER
DECLARE primaryTurns, secondaryTurns AS DOUBLE
DECLARE primaryVoltage, secondaryVoltage, turnsRatio AS DOUBLE
OUTPUT menu (1.Turns Ratio, 2.Secondary V, 3.Primary V)
INPUT choice
SWITCH choice
CASE 1: INPUT N1, N2; turnsRatio = N1/N2; OUTPUT turnsRatio BREAK
CASE 2: INPUT V1, N1, N2; V2 = V1*N2/N1; OUTPUT V2,'V' BREAK
CASE 3: INPUT V2, N1, N2; V1 = V2*N1/N2; OUTPUT V1,'V' BREAK
DEFAULT: OUTPUT 'Invalid choice'
END SWITCH
END
```

3. FLOWCHART (Structured Diagram)

START

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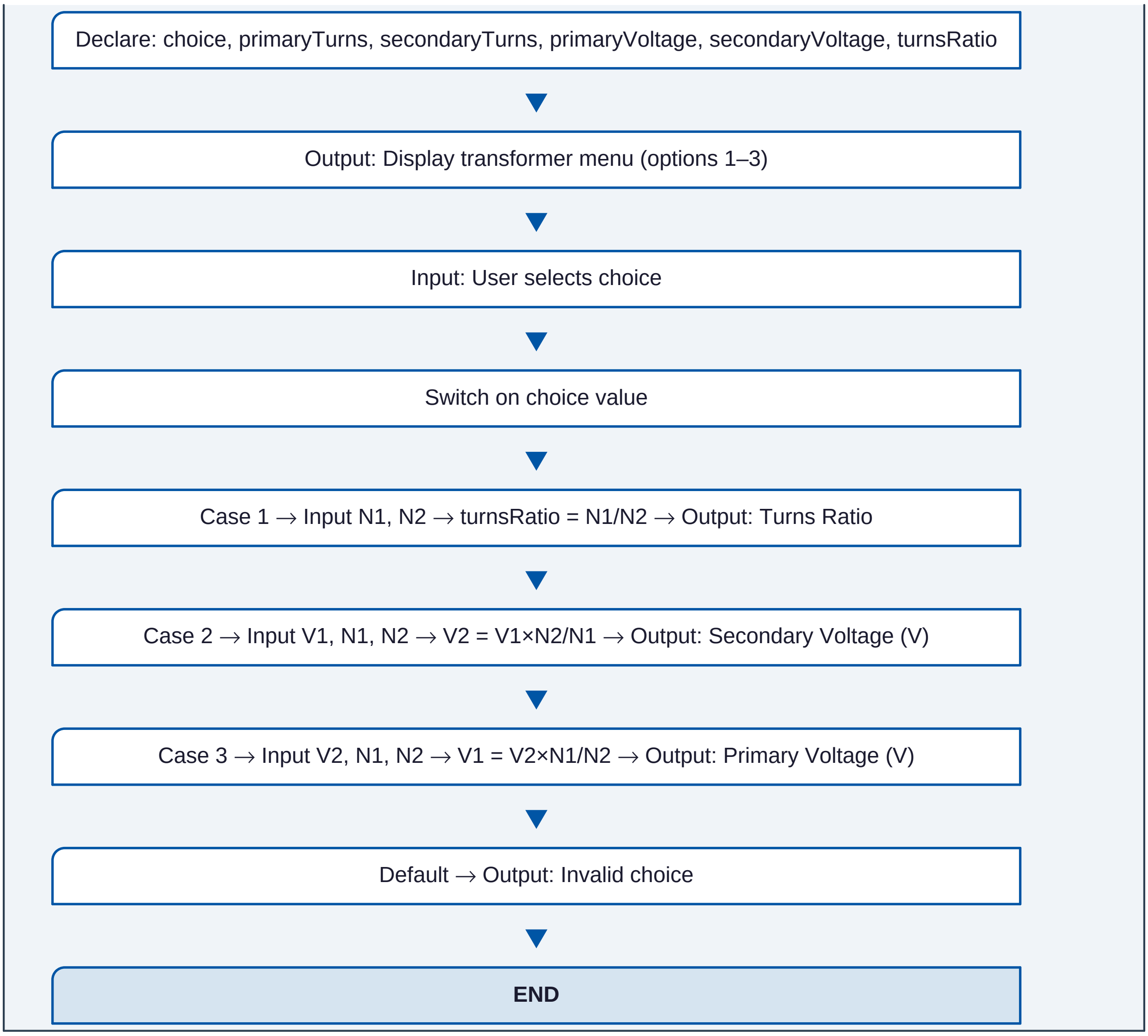

4. TEST RESULTS

Test Item	Value / Description	Test Item	Value / Description
Input	choice=1, N1=200, N2=50	Expected Output	Turns Ratio = 4
Input	choice=2, V1=240V, N1=200, N2=50	Expected Output	V2 = 60 V
Input	choice=3, V2=60V, N1=200, N2=50	Expected Output	V1 = 240 V